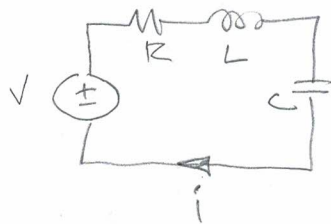


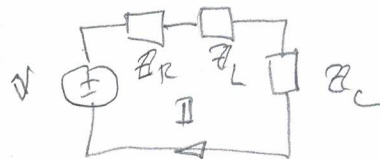
4.54)



$$\begin{cases} V = 12 \cos(3t), \omega = 3 \text{ rad/s} \\ R = 3 \Omega \\ L = 3 \text{ H} \\ C = \frac{1}{3} \text{ F} \end{cases}$$

Bestimmen Sie den Strom $i(t)$ im Kreis.

Lösung: komplex representation



KSPG gilt

$$-V + (Z_R + Z_L + Z_C)I = 0$$

$$\Rightarrow I = \frac{V}{Z_R + Z_L + Z_C} \quad (1)$$

$$\text{Mit } \begin{cases} Z_R = R = 3 \Omega \\ Z_L = j\omega L = j3 \cdot 3 = j9 \Omega \\ Z_C = \frac{1}{j\omega C} = -j \Omega \end{cases} \quad (2)$$

(2) in (1)

$$I = \frac{12 \angle 0^\circ}{3 + j9 - j} = \frac{12 \angle 0^\circ}{3 + j8} = \frac{12 \angle 0^\circ}{8.54 \angle 69.44^\circ} =$$

$$= 1.405 \angle -69.44^\circ$$

Amplitudenwert $|I|$

$$i(t) = \text{Re}(I) = 1.41 \cos(\omega t - 69.4^\circ) \text{ A.}$$

Skizze: $i(t) = 1.41 \cos(\omega t - 69.4^\circ) \text{ A}$